File permissions in Linux

Project description

As a security professional supporting a research team, I was responsible for reviewing and updating file and directory permissions to ensure users had the appropriate level of access. This project involved examining existing permissions, identifying unauthorized access, and modifying permissions to align with organizational security requirements. Using Linux command-line tools, I applied the principle of least privilege to strengthen system security.

Check file and directory details

To review existing permissions for files and directories, I used:

```
ls -la
```

This command displays detailed information about files, directories, hidden files, ownership, and permission settings. The output showed permission for several project files, a hidden file named `.project_x.txt`, and a directory named `drafts`.

Describe the permissions string

Example:

```
-rw-rw-r—- project_k.txt
```

The 10-character string represents the file type and permission settings:

1. Character 1 (–) indicates a regular file.
2. Characters 2–4 (`rw-`) represent user permissions (read and write).
3. Characters 5–7 (`rw-`) represent group permissions (read and write).
4. Characters 8–10 (`r—-`) represent other user permissions (read only).

These permissions determine who can read, write, or execute a file.

Change file permissions

The organization does not allow others to have write access to files. To remove write access for others on `project_k.txt`, I used:

```
chmod o-w project_k.txt
```

This command removed the write permission from the "others" category while preserving existing permissions for the file owner and group.

Change file permissions on a hidden file

The hidden file `.project_k.txt` was archived and should only allow read access for the user and group.

I used:

```
chmod u-w,g+r,g-w.project_x.txt
```

This command removed write access from the user, ensured the group had read access, and removed write access from the group.

Change directory permissions

Only researcher2 should have access to the `drafts` directory and its contents.

I used:

```
chmod g-x drafts
```

This command removed execute access from the group, preventing group members from entering the directory.

Summary

In this project, I examined Linux file and directory permissions using the `ls-la` command and modified permissions using `chmod`. I removed unauthorized write access, secured a hidden file, and restricted access to a sensitive directory. These actions helped enforce the principle of least privilege and improved the overall security of the research team's file system.